

# Study, Modeling and Simulation of a Hybrid Ship Propulsion System

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**Abstract**—In recent years, there has been a great deal of interest in hybrid propulsion systems that provide power improvement solutions for watercraft. In order to be able to exploit the hybrid propulsion systems, one must first understand the principles of operation, its constituents as well as the parameters that influence its production and the means to control them. It is also essential to classify different types of batteries and panels according to their applications and environments. Our study will focus on the use of fuel cell, photovoltaic panels, a generator and a battery in the propulsion of a ship, which will choose an electrical topology that meets the constraints imposed. To realize a good exploitation of the hybrid system it is necessary to study all the equipment and to highlight the parameters on which they depend. Hence, it is necessary to think about the use of various sources to power ship propulsion system and that will allow better performance and strengthen navigation security. In this paper, we firstly studied, series and parallel hybrid propulsion configurations. Then we established the modeling of this system. Finally, we gave a numerical simulation of the proposed hybrid propulsion system operation.

**Keywords**—*Propulsion, hybrid system, estimation of propulsive power, numerical simulation*

## I. INTRODUCTION

Like any floating body capable of moving, the ship must provide some propulsive power to overcome the force opposing its movement, which is "Resistance to Advancement".

Today internal combustion engines are the most used in propulsion despite their low yields and their adverse effects on the environment and the price of oil that does not stop increasing. For this, researchers have used renewable energies on board ships to face the oil cost challenge, to protect the environment and to overcome the problem of ships' dependency to fossil fuel [1]. We will combine various heterogeneous sources of energy together in addition to an energy storage unit in order to satisfy the propulsive power required by the hybrid propulsion system.

The purpose of this article is the study of the hybrid propulsion of a ship using photovoltaics, fuel cell, generator and batteries.

In the first part of this paper we will study the two main hybrid propulsion configurations onboard a ship and mainly hybrid series and parallel hybrid, as well as the various technological variants. The second part gives a study on the ship's hull resistance to the advancement and the estimate of the propulsive power. A third part is dedicated to the modeling of a hybrid propulsion system. Finally we give the numerical simulation results of this hybrid propulsion system.

## II. HYBRID PROPULSION OF A SHIP

Hybridization in the naval domain can manifest itself under several structures, topologies and connection architectures.

The use of hybrid systems is a special recommended approach for ships that requires a high degree of maneuverability and operate on very varied cycles. Like any system, hybrid propulsion has its own advantages and disadvantages compared to traditional fuel-based propulsion. As an advantage, the hybrid propulsion offers more architectural freedom during the design of the ship, it allows a better optimization of the installed power on board and offers a silent propulsion with less vibrations, it gives a more economic propulsion with low CO<sub>2</sub> emission, it ensures more operation flexibility and it improves ship's maneuverability. However, as a disadvantage, the investment cost is higher and hybrid systems technologies are usually more complicated [2].

## III. DIFFERENT STRUCTURES OF HYBRID PROPULSION

### A. Hybrid series

In this type of hybrid propulsion system there are two different energy sources that are used in series mainly the internal combustion engine and electrical systems. Propulsion is provided by the electric motor which is linked to its converter [3, 4].

Among the advantages of hybrid series propulsion: it is easy to control, energy recovery is possible, and little maintenance is needed. Its disadvantage is the low steady-state efficiency and high installation cost [5, 6].

### B. Parallel Hybrid

Parallel hybrid propulsion is an architecture that combines two propulsion engines, an electric motor and a heat engine [4]. Both motors are mechanically linked by the same shaft line through a clutch system and gearbox. The electric motor is powered by the energy storage system and / or by an independent source [5, 6]. As advantages, the parallel hybrid offers a good performance and offers the ability to navigate in electric mode only. The disadvantage resides in the complication of this system in terms of maintenance and installation point of view [6].

### C. Hybrid series-parallel

This combination allows the switch from the parallel hybrid to the hybrid series which gives more flexibility to the propulsion system and ensures a very high redundancy by combining these two modes.

By comparing these different structures, we come to the fact that the structure that offers the most benefit is the series hybrid structure that will be adopted in this work.

## IV. THE DIFFERENT ARCHITECTURES AND TOPOLOGIES OF CONNECTION

### A. DC bus architecture

In the DC bus architecture, the energy generators are connected to a single DC bus that centralizes all the energies produced. The advantage of this topology is the simplicity of the control system by the use of the DC bus, as well as the operation of the generator in the optimal regime.

But the disadvantage is the low efficiency of the system by the intervention of converters and batteries as well as the expensive coupling which is more complex than the AC coupling [5, 6].

### B. AC bus architecture

In the AC bus architecture, the energy generators are connected to a single AC bus, which centralizes all the energies produced [5, 6]. This time the use of converters is intended to supply the AC bus with current as well as to supply each type of load with the appropriate type of current [7]. The advantages are: Simplified photovoltaic connection in AC and simplified power extension. The disadvantages are: the need to over-size loader inverters to provide sufficient load capacity capable of draining power supplied from multiple power sources and the need for programming more complex components.

### C. Mixed architecture

This architecture combines the two architectures mentioned above.

The mixed architecture uses two buses, one bus to DC and the other to AC. The two buses are interconnected by a bidirectional converter or an inverter and a rectifier.

The advantages are: the possibility to feed a part of the load without going through a converter which increases the power system efficiency; besides, the diesel group and the inverter can operate in parallel to ensure load peaks. The disadvantage is the complexity of the system due to the parallel operation of the diesel and inverter unit [5, 6, 7].

## V. SYNOPTIC DIAGRAM OF THE HYBRID PROPULSION SYSTEM

The adopted hybrid electric system includes fuel cells, photovoltaic panels, generators, batteries, static converters, an inverter, asynchronous motor and a fixed blade propeller.

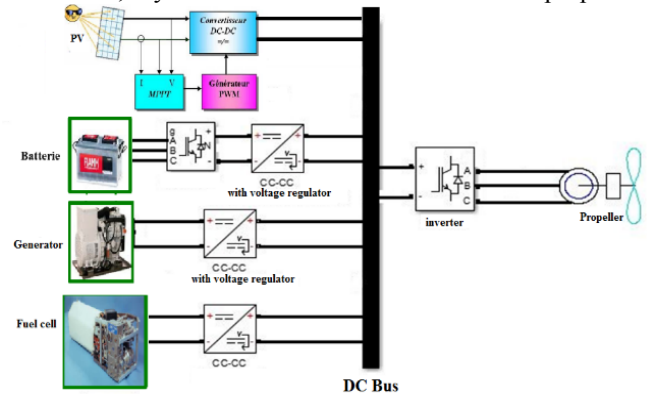


Fig.1. Synoptic diagram of the hybrid propulsion system

## VI. MODELING A HYBRID PROPULSION SYSTEM

### A. Modeling the photovoltaic PV system

The solar cells are generally associated in series and in parallel, then encapsulated under glass to obtain a photovoltaic module. A PV generator consists of interconnected modules to form a unit producing a high continuous power compatible with the usual electrical equipment. PV modules are usually connected in series-parallel to increase the voltage and current at the output of the generator. The interconnected modules are mounted on metal supports and inclined at the desired angle depending on the location, this set is often referred to as module field. Figure 2 shows the curve  $I = f(V)$  of a typical photovoltaic module under constant conditions of irradiation and temperature. The standard irradiation adopted for measuring the response of photovoltaic modules is a radiant intensity of  $1000\text{W/m}^2$  and a temperature of  $25^\circ\text{C}$ .

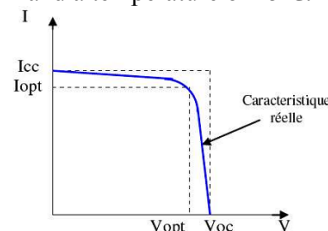


Fig. 2. Characteristic  $I = f(V)$  of a photovoltaic module

The photovoltaic module is considered as a source of power at the point  $P_m$  where the power is maximum. It is therefore interesting to place oneself on this point to get the most energy and thus make the most of the peak power installed.

This task is known in literature as the problem of Maximum Power Point Tracking (MPPT).

Figure 3 shows the influence of illumination on the characteristic  $I=f(V)$ . At a constant temperature, it is found that the current undergoes a significant variation, but against the voltage varies slightly. Because the short-circuit current is a linear function of illumination while the open-circuit voltage is a logarithmic function [7, 11, 13].

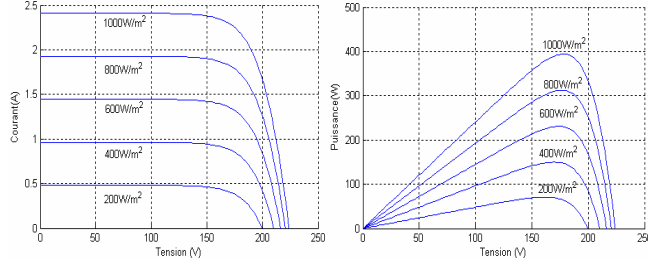


Fig. 3. The influence of illumination on the characteristic  $I = f(V)$  and the characteristic  $P = f(V)$

Figure 4 shows the influence of the temperature on the characteristic  $I = f(V)$  and the power delivered by the generator as a function of the voltage for different values of the temperature. It is essential to understand the effect of varying the temperature of a solar cell on the characteristic  $I = f(V)$ . When the temperature increases the open circuit voltage decreases. As a result, the maximum power of the generator is reduced [11,13].

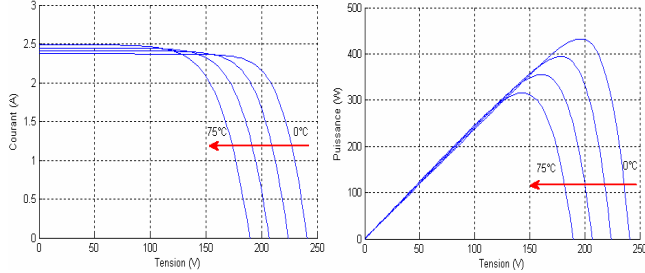


Fig. 4. The influence of temperature on the characteristic  $I=f(V)$  and the characteristic  $P = f(V)$

The I-V characteristic of the PV generator is based on that of an elementary cell modeled by the well-known equivalent circuit of figure 5 [11]. This circuit introduces a current source and a diode in parallel, as well as series resistors  $R_s$  and parallel (shunt) resistors  $R_{sh}$  to take into account dissipative phenomena at the cell.

The equivalent diagram of a photovoltaic cell is as follows:

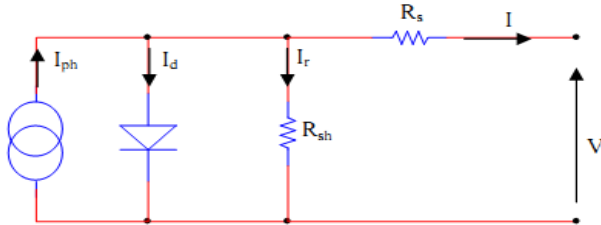


Fig. 5. Equivalent diagram of a photovoltaic cell

The equation connecting the current delivered by a PV module consisting of putting  $N_s$  cells in series and the voltage at its terminals is given by:

$$I = I_{ph} - I_0 \cdot e^{\frac{v + I \cdot R_s}{V_T}} - \frac{V + I \cdot R_s}{R_{sh}} \quad (1)$$

Where:  $I_{ph}$ ,  $I_0$  and  $V_T = \frac{N_s \cdot n \cdot k \cdot T}{q}$  respectively denote

the photocurrent, the inverse saturation current of the diode and the thermal voltage, with:  $n$  the ideality factor of the diode,  $q$  the charge of the electron,  $k$  the Boltzman constant,  $T$  the temperature of the cell which varies according to the illumination and the ambient temperature [9].

### MPPT commands for DC-DC converters

There are several algorithms to find the operating point of the solar module so that the generated power is maximum without interruption of system operation.

The power extracted from the module is calculated from the measurements of current  $I$  and voltage  $V$  of the module and the multiplication of these two quantities.

$$P_{pv} = V_{pv} \cdot I_{pv} \quad (2)$$

These measurements are used by various methods that pursue the real MPP of the range that I chose the inductance increment algorithm where the derivative of the output power of the panel is calculated in another way. It is calculated as a function of the voltage  $V$ , its variation  $dV$ , the current  $I$  and its variation  $dI$ . Power derivative is zero at the point of maximum power, positive to the left of the MPP point and negative to the right [6, 10].

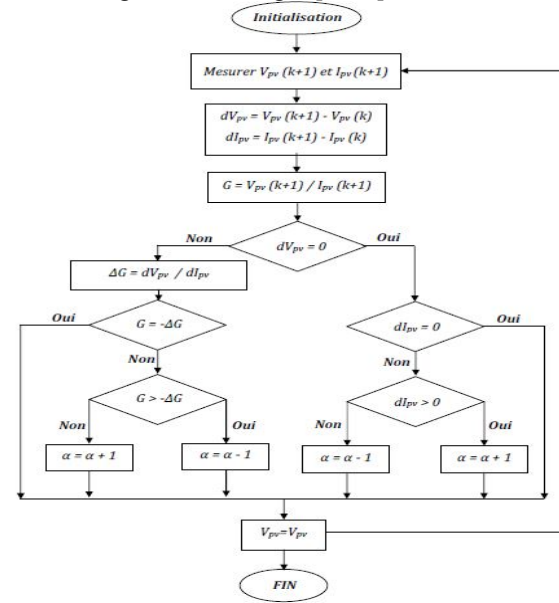


Fig. 6. Flowchart of the INC algorithm

The voltages and currents of the panel are monitored, so that the controller could calculate the conductance and the incremental conductance, and decide on its behavior. The algorithm of Figure 6 involves a large number of derivative computations.

### B. Modeling of the fuel cell:

The voltage of a cell in the stack can be calculated from the following equation [13]:

$$V_{cell} = E - V_{act} - V_{ohm} \quad (3)$$

The electromotive force  $E_{cell}(V)$  could be calculated by (4) [16]

$$E_{cell} = 1.229 - 8.5 \times 10^{-4} (T - 298.15) + \frac{RT}{2F} \left[ \ln(P_{H_2}) + \frac{1}{2} \ln(P_{O_2}) \right] \quad (4)$$

Where T is the temperature (K), R is the perfect gaz constant, F is Faraday Constant ( $F=96485$  C/mol),  $P_{H_2}$  and  $P_{O_2}$  are the partial pressure of oxygene and hydrogen respectively.

### C. The generator:

In order to simplify the calculations and to facilitate the interpretation of the results, a unitary system was used to standardize all electrical quantities. By applying the Park transform, the equations of the stator and rotor voltages along the d and q axes are written [14]:

$$V_d = -R_a i_a + \frac{d\Psi_d}{\omega_n dt} - \Omega_r \Psi_q \quad (5)$$

$$V_q = -R_a i_q + \frac{d\Psi_q}{\omega_n dt} + \Omega_r \Psi_d \quad (6)$$

$$V_f = R_f i_f + \frac{d\Psi_f}{\omega_n dt} \quad (7)$$

$$0 = R_D i_D + \frac{d\Psi_D}{\omega_n dt} \quad (8)$$

$$0 = R_Q i_Q + \frac{d\Psi_Q}{\omega_n dt} \quad (9)$$

Where :

$\omega_n$  : Reference pulse (rd/s)

$\Omega_r$  : Mechanical pulse in p.u

$R_a$  : Resistance of a stator phase in p.u

$R_f$  : Resistance of the inductor in p.u

$R_D, R_Q$  : resistance of damping windings in p.u.

$i_a$  : Proper Inductance of the inductor in p.u

$i_q$  : Proper Inductance of damping windings

$i_f$  : Excitation inductance

$i_D$  and  $i_Q$  : own inductance of the dampers in p.u

$\Psi_d, \Psi_q$  and  $\Psi_f$  : are flow through the windings d, q in p.u

$m_{af}, m_{fd}, m_{aQ}, m_{aD}$  : mutual inductances between coils in p.u.

$u_f$  : Excitation voltage in p.u.

$u$  : Armature voltage in p.u

$P$  : Active power in p.u

$Q$  : Reactive power in p.u.

The shaft dynamics equation of the diesel engine:

$$J_m \Omega_r = C_m - C_e \quad (10)$$

With:  $J_m$ : moment of inertia of the shaft line,  $C_e$ : electromagnetic torque,  $C_m$ : mechanical torque of the diesel engine,  $\omega_n$ : Reference pulsation (rd/s),  $\Omega_r$ : Mechanical pulsation in p.u

The expression of the electromagnetic torque is as follow:

$$C_e = \Psi_d i_q - \Psi_q i_d \quad (11)$$

$\Psi_f, \Psi_d, \Psi_q$  : are flow through the windings d, q in p.u and  $i_d$ ,

$i_q$ : inductance of the dampers in p.u

The expressions of active and reactive powers are:

$$P = u_d i_d + u_q i_q \quad (12)$$

$$Q = u_q i_d - u_d i_q \quad (13)$$

Where  $u$  : armature voltage in p.u

### D. The batteries

Several battery models found in literature use a simple model as shown in Figure 7. It includes an electromotive force  $E_0$ , modeling the empty battery voltage, a capacitor  $C_{bat}$ , modeling the internal capacity of the battery and an internal resistance  $R_{bat}$  [17].

So we have :

$$V_{bat} = E_0 - R_{bat} \cdot i - V_c \quad (14)$$

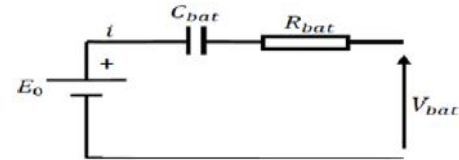


Fig. 7. Electric model of the battery

### E. Model of the three-phase inverter

The variable speed drive consists of a PWM three-phase inverter [6].

$$\begin{pmatrix} V_{AN} \\ V_{BN} \\ V_{CN} \end{pmatrix} = \frac{V_{bus}}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \cdot \begin{pmatrix} S_1 \\ S_2 \\ S_3 \end{pmatrix} \quad (15)$$

where  $S_1, S_2$  and  $S_3$  denote the states of the switches of the different phases A, B and C respectively and  $V_{bus} = 400$  V.

### F. Asynchronous motor model with permanent magnet

The MSAP is brought back to a machine with a pair of poles, the angle  $\theta_r$  will correspond to the real angle of the rotor multiplied by the number of pairs of poles P.

If we consider a sinusoidal distribution of the magnetic induction and neglecting the saturation phenomenon we will have in the reference d-q the following equations:

#### Voltage equations

$$V_d = R_s I_d + \frac{d\phi_d}{dt} - \omega_r \phi_q \quad (16)$$

$$V_q = R_s I_q + \frac{d\phi_q}{dt} + \omega_r \phi_d \quad (17)$$

#### Flows equations

$$\phi_d = L_d I_d + \phi_f \quad (18)$$

$$\phi_q = L_q I_q \quad (19)$$

where  $L_d(H)$  and  $L_q(H)$  are the equivalent armature inductance on the d and q axis, respectively;  $R_s(\Omega)$  is the equivalent resistance of stator winding; P is the number of pole pairs; F is the coefficient of fluid friction and J is the inertia of the rotor.

Taking as state vector the two components of the current on the d and q axes for input vector equations (1.1) and (1.4) makes it possible to obtain the following equation of state.

$$\begin{pmatrix} \dot{I}_d \\ \dot{I}_q \end{pmatrix} = \begin{pmatrix} -\frac{R_s}{L_d} & \frac{L_q}{L_d} \omega_r \\ -\frac{L_d}{L_q} \omega_r & -\frac{R_s}{L_q} \end{pmatrix} \begin{pmatrix} I_d \\ I_q \end{pmatrix} + \begin{pmatrix} \frac{1}{L_d} & 0 & 0 \\ 0 & \frac{1}{L_q} & -\frac{\omega_r}{L_q} \end{pmatrix} \begin{pmatrix} V_d \\ V_q \\ \phi_f \end{pmatrix} \quad (20)$$

### G. Determination of the resistance to advancement and estimation of the propulsive power

The evaluation of the resistance to the advancement is one of the essential phases in the conduct of a ship design project since it conditions directly the dimensioning of the propulsion apparatus and the operational capacities of the ship. Several studies are carried out to determine the resistance applied to the ship during its displacement. During its movement, the ship is subjected to a set of resistance forces to progress that will affect its speed and the propulsion machine that will be installed on board according to its characteristics. The total resistance is given in Newton and it is estimated by the expression (21) [11, 12]:

$$R_T = [R_F(1 + K_1) + R_W + R_{APP} + R_B + R_{TR} + R_A + R_{AIR}] \times (1.0 + DMAR / 100) \sqrt{a^2 + b^2} \quad (21)$$

Where:

$R_T$ : Total resistance to movement

$R_F$ : friction resistance,

$1 + K_1$ : coefficient depending on the hull's shape,

$R_{APP}$ : appendages resistance,

$R_W$ : wave resistance,

$R_B$ : bulbous resistance,

$R_A$ : resistance of the roughness and air resistance,

$R_{TR}$ : Whirlpool resistance can be neglected because the hull's shape,

$R_{AIR}$ : aerodynamic drag,

DMAR: refers to the margin on the strength in percent. The total resistance is increased by the amount (+1.0 DMAR/100).

The general formula for calculating the towing power is given by the following relation (22) [11], [12]:

$$P_{rem} = R_T \times V \quad (22)$$

The useful power is given by the following relations [11], [5]:

$$P_{util} = \frac{P_{rem}}{\eta_g} \quad (23)$$

With  $\eta_g$  the overall yield which is expressed as:

$$\eta_g = \eta_c \eta_A \eta_H \eta_M \quad (24)$$

where:

$\eta_c$ : Efficiency of the hull,  $\eta_A$ : Efficiency of helix adaptation to the hull,  $\eta_H$ : Efficiency of the helix,  $\eta_M$ : Mechanical yield.

### H. Global model of the propulsion system

The overall model of the propulsion system is a multivariate nonlinear system of order 4 given by:

$$\begin{cases} \frac{di_d}{dt} = -\frac{R_s}{L_d} i_d + p \Omega \frac{L_q}{L_d} i_q + \frac{1}{L_d} u_d \\ \frac{di_q}{dt} = -p \Omega \frac{L_d}{L_q} i_d - \frac{R_s}{L_q} i_q - \frac{p \phi_f}{L_q} \Omega + \frac{1}{L_q} u_q \\ \dot{\Omega} = \frac{1}{I_m} \left[ \left( p(L_d - L_q) i_d i_q + p \phi_f i_q \right) - \frac{1}{4\pi^2} (\rho \Omega^2 D^5 s_1) \right] \\ \dot{V} = \frac{1}{m} \left[ -aV^2 + \frac{1}{4\pi^2} (1-t) \eta \rho D^4 \Omega^2 + \frac{(1-t)(1-w)}{2\pi} r_2 \rho D^3 \Omega V - T_{ext} \right] \end{cases} \quad (25)$$

Where:  $\rho$ : Seawater density,  $D$ : Diameter of the thruster,  $n$ : Rotation speed of the propeller,  $V_a$ : advance speed of the propeller,  $T$ : the propulsive force (pushed),  $Q$ : the propeller torque,  $K_Q$  and  $K_T$  are the propulsion coefficient.

## VII. ENERGY MANAGEMENT SYSTEM

Since our propulsion system is multi-source, it will be necessary to have a power management system capable of managing the system in real time. For that we proposed the following algorithm:

**Algorithm:**

$$SoC_{min} = 20$$

$$SoC_{max} = 100$$

$$OP = \begin{cases} 1 & \text{if generator is ON} \\ 0 & \text{if generator is OFF} \end{cases}$$

$$P_{bus}(k) = P_{pv}(k)$$

$$R_T(k) = f(V(k))$$

$$P_{load}(k) = R_T(k) \cdot V(k)$$

if  $P_{bus}(k) > P_{load}(k)$  then

if  $SoC < SoC_{max}$  then

$$P_{bus}(k+1) = P_{pv}(k+1) - P_{bat}(k+1)$$

else  $P_{bus}(k+1) = P_{pv}(k+1)$

if  $P_{bus}(k) = P_{load}(k)$  then

$$P_{bus}(k+1) = P_{pv}(k+1)$$

if  $P_{bus}(k) < P_{load}(k)$  then

if  $OP = 1$  then

if  $SoC < SoC_{max}$  then

$$P_{bus}(k+1) = P_{pv}(k+1) + P_{ge}(k+1) - P_{bat}(k+1)$$

else  $P_{bus}(k+1) = P_{pv}(k+1) + P_{ge}(k+1)$

if  $OP = 0$  then

if  $SoC > SoC_{min}$  then

$$P_{bus}(k+1) = P_{pv}(k+1) + P_{bat}(k+1)$$

else

$$P_{bus}(k+1) = P_{pv}(k+1) + P_{pac}(k+1)$$

Where:

$k$ : Incrementing variable.

OP: Boolean variable that defines the operational status of the generator.

$P_{bus}$ : Available power on the bus.

$P_{load}$ : Power required to propel at the set speed.

$P_{pv}$ : Photovoltaic power.

$P_{ge}$ : Power of the generator.

$P_{pac}$ : Power of the fuel cell.

$P_{bat}$ : Battery power.

$V$ : desired speed

$R_T$ : Total resistance to progress corresponds to the desired speed.

$SoC$ : Situation of charge of the battery.

### Estimation of the propulsive power:

In this part, the propulsive power was estimated using the Power Prediction Program (PPP) software. The sailboat is equipped with a *Volvo Penta* type diesel propulsion engine of power  $P = 27$  hp; therefore  $P = 19.9$  kW operating at the speed of 8 Knots. So, according to the table shown in figure 8 and using a simple interpolation, we can determine the maximum speed with the old propulsion mode. Hence, we find  $V_{max} = 8.44$  Knots. For our project propulsion system, we propose a permanent magnet synchronous electric motor of power  $P = 18$  kW.

### VIII. SIMULATION OF THE HYBRID PROPULSION SYSTEM

The results obtained in figure 8 are for ship's speed ranging from 1 to 8 knots with an increment of 1 knot.

| V(kts) | Form<br>RF*K1 | Appendage<br>RAPP | Wave<br>RW | Bulb<br>RB | Transom<br>RTR | Correlation<br>RA | Air<br>RAIR | Dreg<br>RAIR |
|--------|---------------|-------------------|------------|------------|----------------|-------------------|-------------|--------------|
| 1.00   | 4.3           | 0.2               | 0.0        | 0.0        | 0.0            | 3.6               | 0.0         | 0.0          |
| 2.00   | 15.3          | 0.7               | 0.0        | 0.0        | 0.0            | 14.6              | 0.0         | 0.0          |
| 3.00   | 32.2          | 1.5               | 0.2        | 0.0        | 0.0            | 32.8              | 0.0         | 0.0          |
| 4.00   | 64.5          | 2.7               | 7.6        | 0.0        | 0.0            | 58.3              | 0.0         | 0.0          |
| 5.00   | 82.2          | 4.3               | 68.0       | 0.0        | 0.0            | 91.1              | 0.0         | 0.0          |
| 6.00   | 114.9         | 6.1               | 329.2      | 0.0        | 0.0            | 131.2             | 0.0         | 0.0          |
| 7.00   | 152.7         | 8.4               | 794.3      | 0.0        | 0.0            | 178.6             | 0.0         | 0.0          |
| 8.00   | 195.3         | 10.9              | 1728.2     | 0.0        | 0.0            | 233.3             | 0.0         | 0.0          |

Resistance, Effective Power, Propulsion Factors and Required Thrust

| V(kts) | RT(N)  | PE(kW) | w      | t      | REQ. THR(N) | etaH   | etaRR  |
|--------|--------|--------|--------|--------|-------------|--------|--------|
| 1.00   | 24.8   | 0.01   | 0.5758 | 0.2903 | 34.9        | 1.6729 | 1.0042 |
| 2.00   | 89.5   | 0.09   | 0.5281 | 0.2903 | 126.1       | 1.5038 | 1.0042 |
| 3.00   | 190.6  | 0.29   | 0.5053 | 0.2903 | 268.5       | 1.4346 | 1.0042 |
| 4.00   | 333.4  | 0.69   | 0.4910 | 0.2903 | 469.8       | 1.3944 | 1.0042 |
| 5.00   | 564.3  | 1.45   | 0.4809 | 0.2903 | 795.1       | 1.3671 | 1.0042 |
| 6.00   | 1034.5 | 3.19   | 0.4731 | 0.2903 | 1457.6      | 1.3470 | 1.0042 |
| 7.00   | 1746.6 | 6.29   | 0.4669 | 0.2903 | 2461.0      | 1.3313 | 1.0042 |
| 8.00   | 2973.0 | 12.24  | 0.4618 | 0.2903 | 4189.1      | 1.3186 | 1.0042 |

Design Margin Has Been Included in RT, PE, and REQ. THR = RT/(1-t).

Fig. 8. Table of calculation results of the propulsive power estimates (ship speed: 1 to 8 kts)

Figure 9 illustrates the resistance to advancement of the ship in addition to an estimation of the required propulsive power. It is to mention that these two values highly increase with ship's speed.

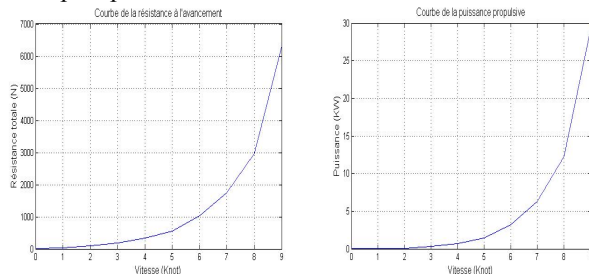


Fig. 9. Curve of estimated total resistance to advancement and estimated propulsive power

The curve in Figure 10 shows the sun illumination for a 24 hour period chosen to simulate the operation of our propulsion system during a day. While, Figure 11 represents the voltage output from the PV panel which depends on sun illumination with a magnitude around 400 V. The current generated from the photovoltaic group is shown in Figure 12. However, Figure 13 displays the shape of the current at the output of the booster chopper. The resulting power generated from the photovoltaic group is presented in Figure 14. It is worthy to notice that the variation in PV power generation is very similar to the illumination curve.

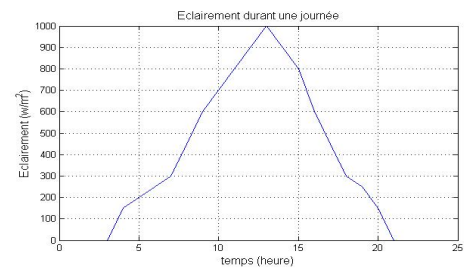


Fig. 10. Curve of sun illumination for 24 hours

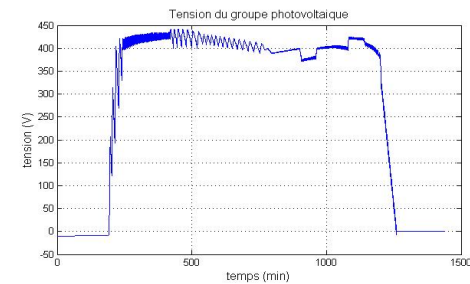


Fig. 11. PV array voltage curve

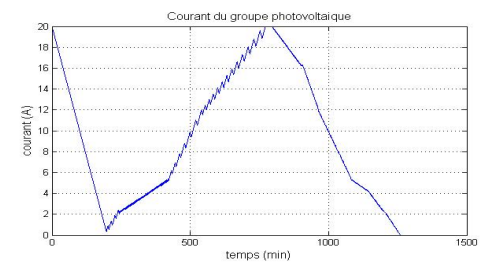


Fig. 12. Current curve of the photovoltaic group

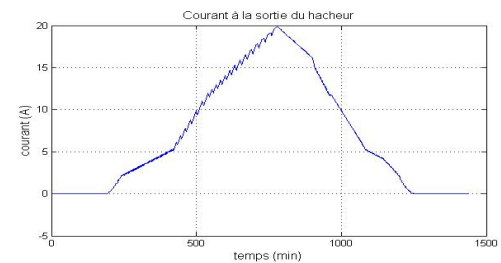


Fig. 13. Current Curve of the Booster Chopper

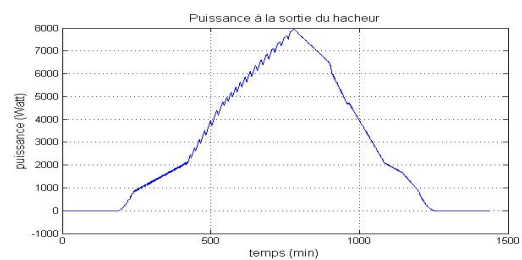


Fig. 14. Power curve of the photovoltaic group

Figure 15 and Figure 16 provides, respectively, a close zoom on the voltage and the current delivered by the alternator.

Figure 17 and Figure 18 illustrate the voltage and the current after rectification, respectively. From Figure 18, we notice that the current still present some fluctuations that could be eliminated by increasing the inductance of the self.

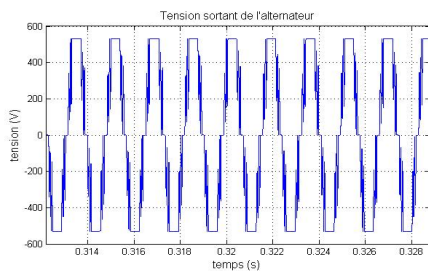


Fig. 15. Zoom on the voltage of the alternator

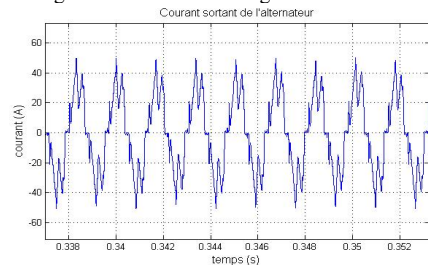


Fig. 16. Zoom on the alternator current

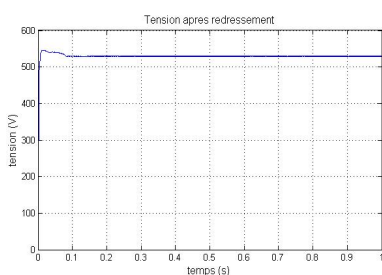


Fig. 17. Voltage curve after rectification

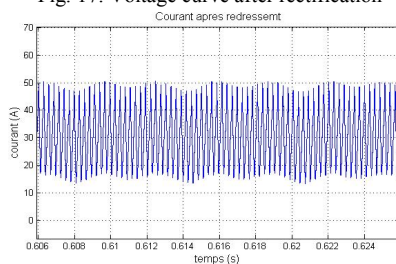


Fig. 18. Curve of the current after rectification

Figure 19 and Figure 20 illustrate, respectively, the curve of the voltage and out-current from the buck converter. The voltage output from the buck converter is around 400V.

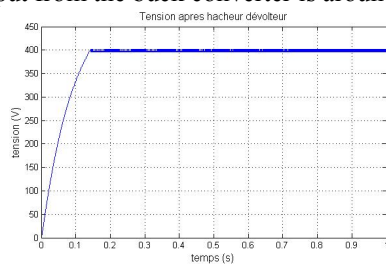


Fig. 19. Voltage curve after buck converter

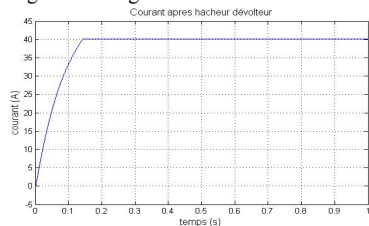


Fig. 20. Current curve after buck converter

Figure 21 depicts the curve of the power produced by the generator. The shape of the current and the voltage at the output of the fuel cell are given by Figure 22 and Figure 23, respectively. Figure 24 and Figure 25 depict, respectively, the current and the voltage output from the boost converter. The power generated by the fuel cell is shown in Figure 26.

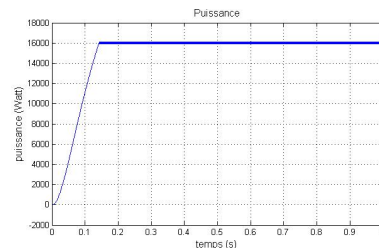


Fig. 21. Generator Power Curve

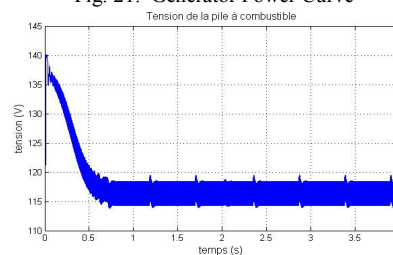


Fig. 22. Voltage curve of the fuel cell

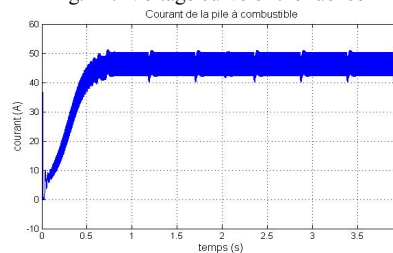


Fig. 23. Current Curve of the Fuel Cell

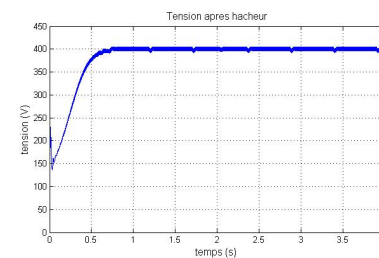


Fig. 24. Voltage output of the boost converter

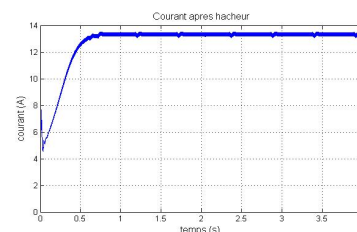


Fig. 25. Current output of the boost converter

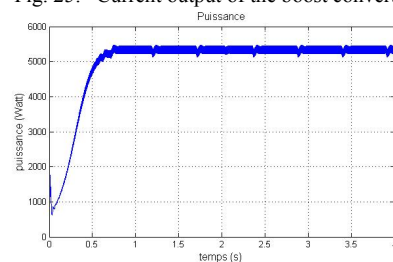


Fig. 26. Power curve of the fuel cell

## IX. CONCLUSION

In this paper we studied hybrid propulsion mainly the hybrid series and the parallel hybrid, as well as the different topologies of ship power system connection. Then, we studied the hydrodynamics of the ships in order to properly estimate the resistance to the advancement and to calculate the required propulsive power in order to properly size the propulsion system. Thereafter, we modeled our propulsion system and we were an application on a ship of known characteristics. Finally, we simulated the hybrid propulsion system of the ship and we checked the proper functioning of each subsystem. The results obtained show a good agreement with the expected performances required to propel the ship.

As future work, we suggest to integrate a supercapacitor in the ship power system in order to improve system performance during motor startups. Besides, the performance of the overall power system should be re-evaluated.

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